CVPR 2023 · NEURAL FIELDS

Canonical Fields

Self-Supervised Learning of Pose-Canonicalized Neural Fields

Rohith Agaram, Shaurya Dewan, Rahul Sajnani,
Adrien Poulenard, Madhava Krishna, Srinath Sridhar

IIIT-Hyderabad • Brown University • Stanford University



Figure 1. We present Canonical Field Network (**CaFi-Net**), a self-supervised method for 3D position and orientation (pose) canonicalization of objects represented as neural fields. We specifically focus on neural radiance fields (NeRFs) fitted to raw RGB images of arbitrarily posed objects (left), and obtain a **canonical field** (fixed novel view shown on right) where all objects in a category are consistently positioned and oriented.

Figure 1 (paper, p. 1): Input NeRFs → Canonical Fields

OUTLINE

The talk in five movements

From the pose problem to a canonical field.

01 **Motivation**
Neural fields & the pose problem

02 **Method**
CaFi-Net architecture & canonicalization algorithm

03 **Data**
1300 NeRF models across 13 ShapeNet categories

04 **Results**
Quantitative comparison, ablations, qualitative analysis

05 **Takeaways**
Contributions, limitations, future work

INTRODUCTION

Neural Fields & the Pose Problem

WHAT THEY ARE

Neural fields encode shape & appearance

Coordinate-based implicit neural networks — the backbone of novel view synthesis, geometry extraction, dynamic scenes, segmentation, and robotics.

WHY POSE MATTERS

Category learning needs one shared pose

Prior methods relied on manually canonicalized datasets (ShapeNet, ModelNet40) or overfitting on a single instance — strong supervision that limits real-world use.

THE GAP

No self-supervised canonicalization operates on neural fields.

Fields are continuous, noisy, and implicit — parameterized as network weights, so direct transformation estimation is challenging. Existing self-supervised methods work only on point clouds, meshes, or voxels.

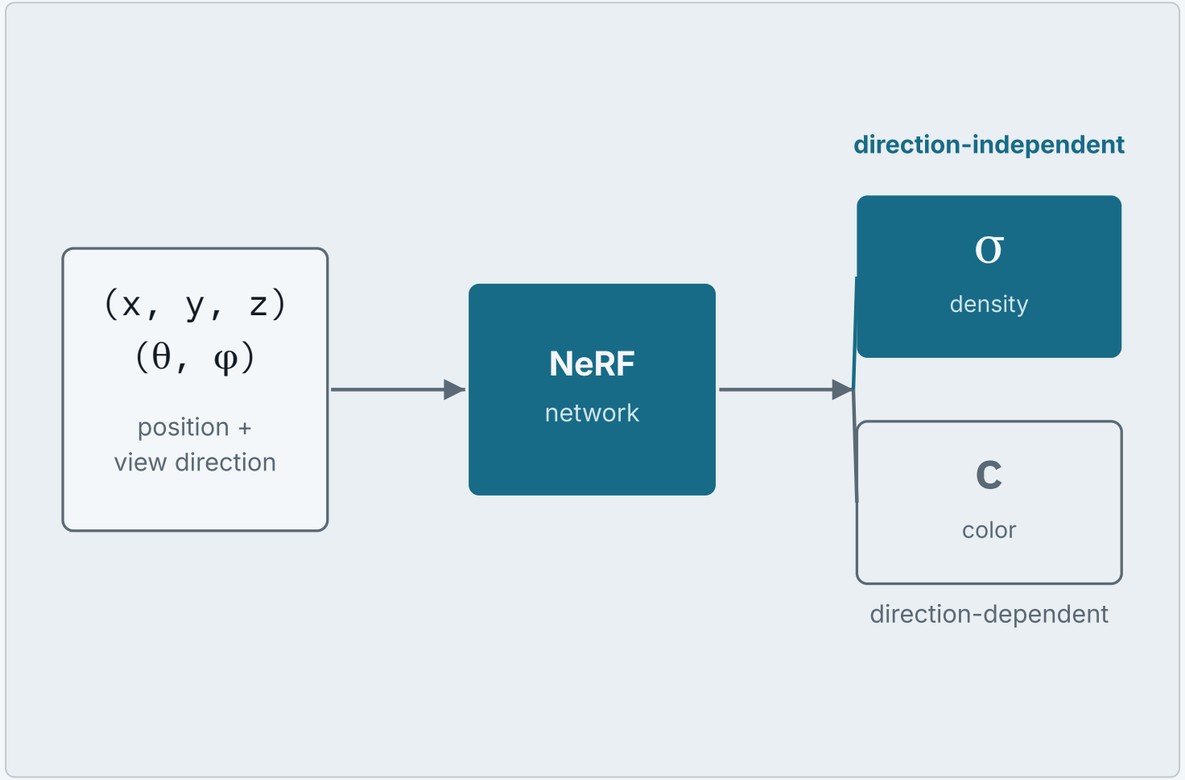
Source: Canonical Fields (CVPR 2023), pp. 1–2

BACKGROUND

NeRFs, Equivariance & Tensor Fields

NERF

From posed images to a density field



Trained on posed RGB images. CaFi-Net uses **density σ** only — color c varies across instances and is discarded.

Source: Canonical Fields (CVPR 2023), pp. 2–3

$$\Gamma(Ox) = M(O)\Gamma(x)$$

Equivariance: features transform consistently with the input. Focus on SE(3), particularly SO(3); translation equivariance via mean centering.

TFN — Tensor Field Networks

Features as type-ℓ spherical harmonics; rotation equivariant by construction.

EQConv — Equivariant Convolution

Aggregates type-ℓ features along receptive fields with learnable weights.

MOTIVATION

Limitations of Existing Methods



Figure 1. We present Canonical Field Network (**CaFi-Net**), a self-supervised method for 3D position and orientation (pose) canonicalization of objects represented as neural fields. We specifically focus on neural radiance fields (NeRFs) fitted to raw RGB images of arbitrarily posed objects (left), and obtain a **canonical field** (fixed novel view shown on right) where all objects in a category are consistently positioned and oriented.

Figure 1 (p. 1) — Input NeRFs at arbitrary poses (left) vs. canonical fields with consistent orientation (right). The goal every method must reach.

Discrete representations only

Recent self-supervised canonicalizers operate on point clouds, meshes, or voxels — **none on neural fields**.

Noisy continuous fields

Fields are continuous, noisy, and implicit; parameterized as network weights, so **direct transformation estimation is challenging**.

Manual canonicalization required

Prior category-level methods overfit a single instance or learn on manually canonicalized datasets (ShapeNet, ModelNet40) — strong supervision limits real-world application.

Source: Canonical Fields (CVPR 2023), pp. 1–2; Figure 1, p. 1

METHOD

Overview of Proposed Method

01

NeRF Sampling

Uniformly sample the density field; find object center & bounding box; normalize density to [0,1]. Color is not used.

$$\sigma_D \in [0,1]$$

→

02

Signal Representation

Three signals per sample: density value, xyz location, and gradient of density.

$$\nabla \sigma_D$$

→

03

Equivariant Convolution

Hierarchical convolutions at resolutions 1/2, 1/4, 1/8 with density-weighted features; global type-ℓ features.

$$F^{\ell}$$

→

04

Canonicalization

Pose-invariant embedding $H^{\ell} \rightarrow$ predicted canonical coordinates P; Siamese branch for a second instance.

$$P$$

Siamese: two instances of the same category are forced into the same canonical frame.

Source: Canonical Fields (CVPR 2023), pp. 3–5

METHODOLOGY

CaFi-Net Architecture

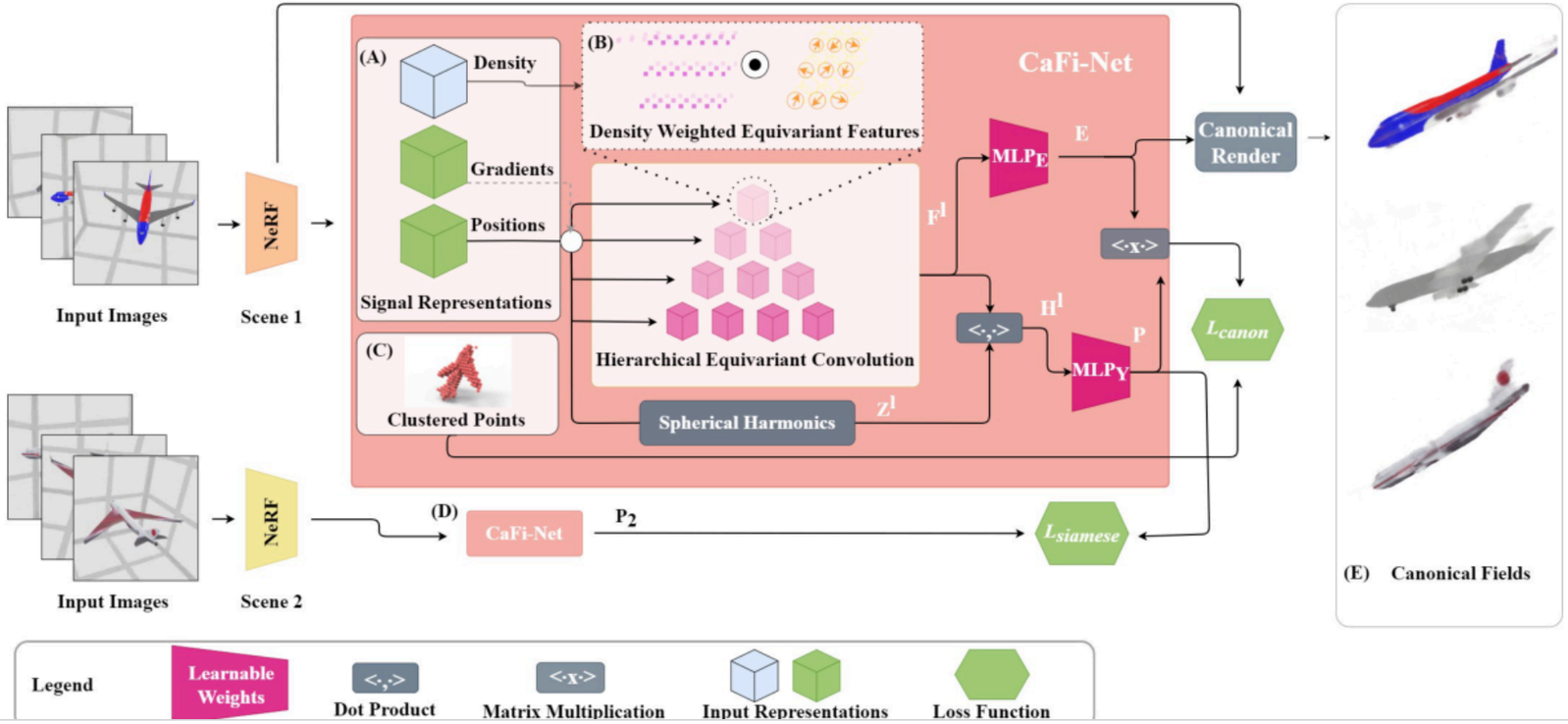


Figure 3 — CaFi-Net: signal representations → hierarchical equivariant convolution → density-weighted features → canonical render.

01

Three signal representations

Density value, xyz location, and gradient of density — gradients capture object surfaces.

02

Hierarchical EQConv

512 neighbors per layer; resolutions 1/2, 1/4, 1/8; features weighted by density.

03

Siamese lower branch

A second instance of the same category, canonicalized into the same canonical frame.

Source: Canonical Fields (CVPR 2023), p. 4, Figure 3

KEY ALGORITHM

Canonicalization & Losses

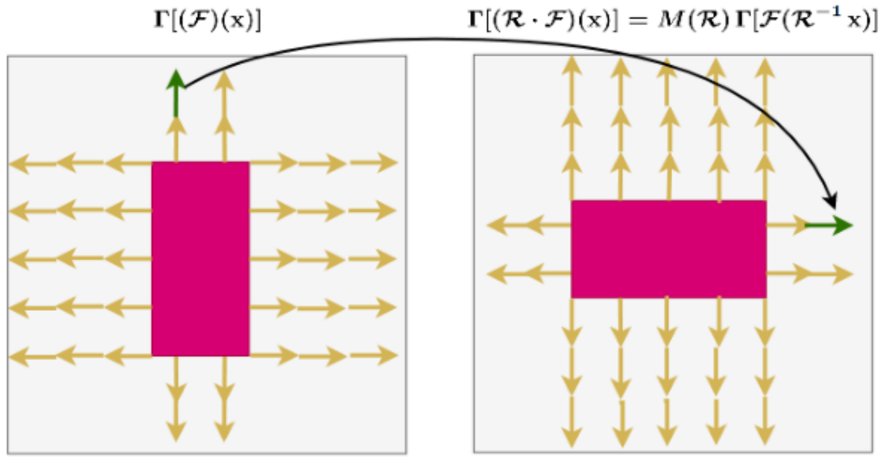


Figure 2. **Rotation Equivariance in Vector Fields.** Rotation equivariance in vector fields $\mathcal{F}$ (here, a simulated 2D magnetic field) require two operations: (1) update the position of the green arrow (the vector), and (2) rotating the green arrow at its new position. A function Γ operating on $\mathcal{F}$ is rotationally equivariant over

Figure 2 (p. 3) — rotation equivariance in vector fields

Two operations: update position, rotate vector — the equivariant convolution preserves this structure across the field.

CANONICALIZATION

$$H^{\ell}(\sigma_D,\ X)\ :=\ \langle F^{\ell}(\sigma_D),\ Z^{\ell}(X)\rangle \tag{4}$$

$$P\ :=\ \mathrm{MLP}_Y(H(\sigma_D,\ X)) \tag{5}$$

$$E\ :=\ \mathrm{MLP}_E(F(\sigma_D,\ X)) \tag{6}$$

Dot product of global equivariant features with point-wise spherical harmonics → **pose-invariant embedding**.

LOSSES

L_{canon} canonicalization	$\mathrm{mean}_{i\in C_f}\ \lVert X_i - E_b\ P_i \rVert_2^2$	**2.0** weight
L_{ortho} orthonormality	$(1/M)\ \Sigma_j\ \lVert E_j - U_j\ V_j^{\top} \rVert_2$	**1.0** weight
$L_{siamese}$ shape consistency	$\mathrm{CD}(P(\sigma_D{}^1),\ P(\sigma_D{}^2))$	**1.0** weight

Foreground cluster C_f: K-Means (K=2) on densities; higher-mean cluster. Eqs. 7–11, p. 5.

Source: Canonical Fields (CVPR 2023), pp. 3, 5, Figures 2–3

DATASET & TRAINING

Dataset Scale & Training Recipe

1300
NeRF models
100 per category

13
ShapeNet categories
randomly rotated

54
views per shape
omnidirectional, Blender; cluttered backgrounds

20%
held out for testing
per category

NeRF Training

Public PyTorch NeRF implementation — fixed epochs for fair comparison

Epochs	**400**
Rays / iteration	**1024 (random)**
Coarse samples / ray	**64**
Fine samples / ray	**128**

Released: simulator, raw posed images, trained NeRF weights

CaFi-Net Training

PyTorch — single GPU, 300 epochs

Epochs	**300**
Batch size	**2**
Hardware	**Nvidia 1080-Ti**
Optimizer	**Adam, lr 6e-4**
L2 regularization	**1e-5**
Loss weights	**L_canon 2.0, others 1.0**

Source: Canonical Fields (CVPR 2023), pp. 5–6

EXPERIMENTAL SETUP

Three Metrics, One Fair Protocol

All metrics are Chamfer-distance based — **lower is better (↓)**.

IC ↓

Instance-Level Consistency — how consistently the **same instance** is canonicalized across different poses.



$P_i, P_j \to C(P_i) \approx C(P_j)$

CC ↓

Category-Level Consistency — how consistently **different instances** of the same category are canonicalized.



$C(P_i) \approx C(P_j),\ i \neq j$

GEC ↓

Ground Truth Equivariance Consistency — fixes ConDor's degenerate GC: rotate P_i by R_1, P_j by R_2, then apply their canonicalizing transforms to P_k.



Eq. 12 · CD = Chamfer distance

BASELINES — ALL POINT-CLOUD (P)

PCA • Canonical Capsules (CaCa) • ConDor

FAIR COMPARISON PROTOCOL

Baselines trained on points sampled from NeRF scenes: 120–1200 points on a 32^3 grid, resampled to ≥1024. **CaFi-Net** estimates the transform from the density field, then transforms its ShapeNet point cloud.

Source: Canonical Fields (CVPR 2023), pp. 5–7

RESULTS

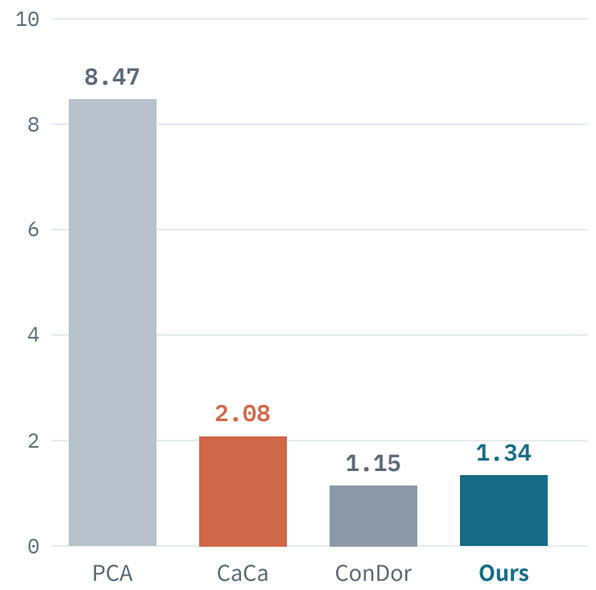
Quantitative Comparison

PCA CaCa ConDor Ours (CaFi-Net)

All metrics ×100, lower is better ↓

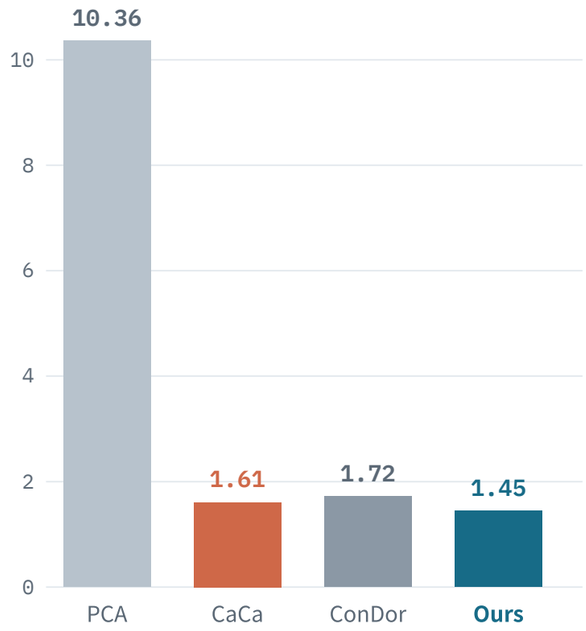
IC

Instance-Level Consistency



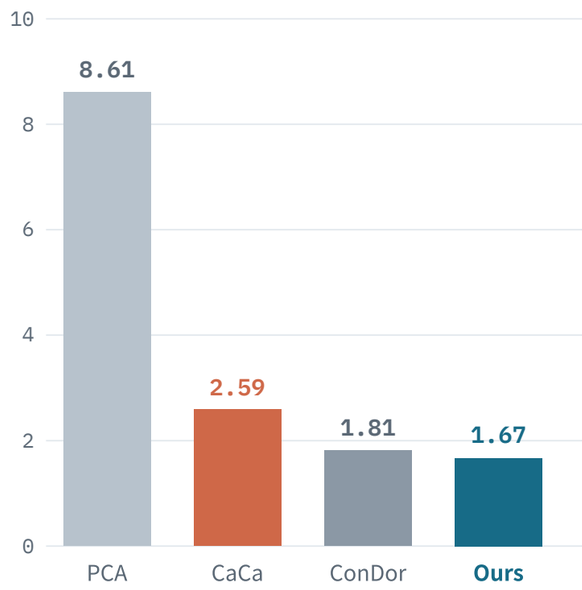
CC

Category-Level Consistency



GEC

Ground-Truth Equivariance Consistency



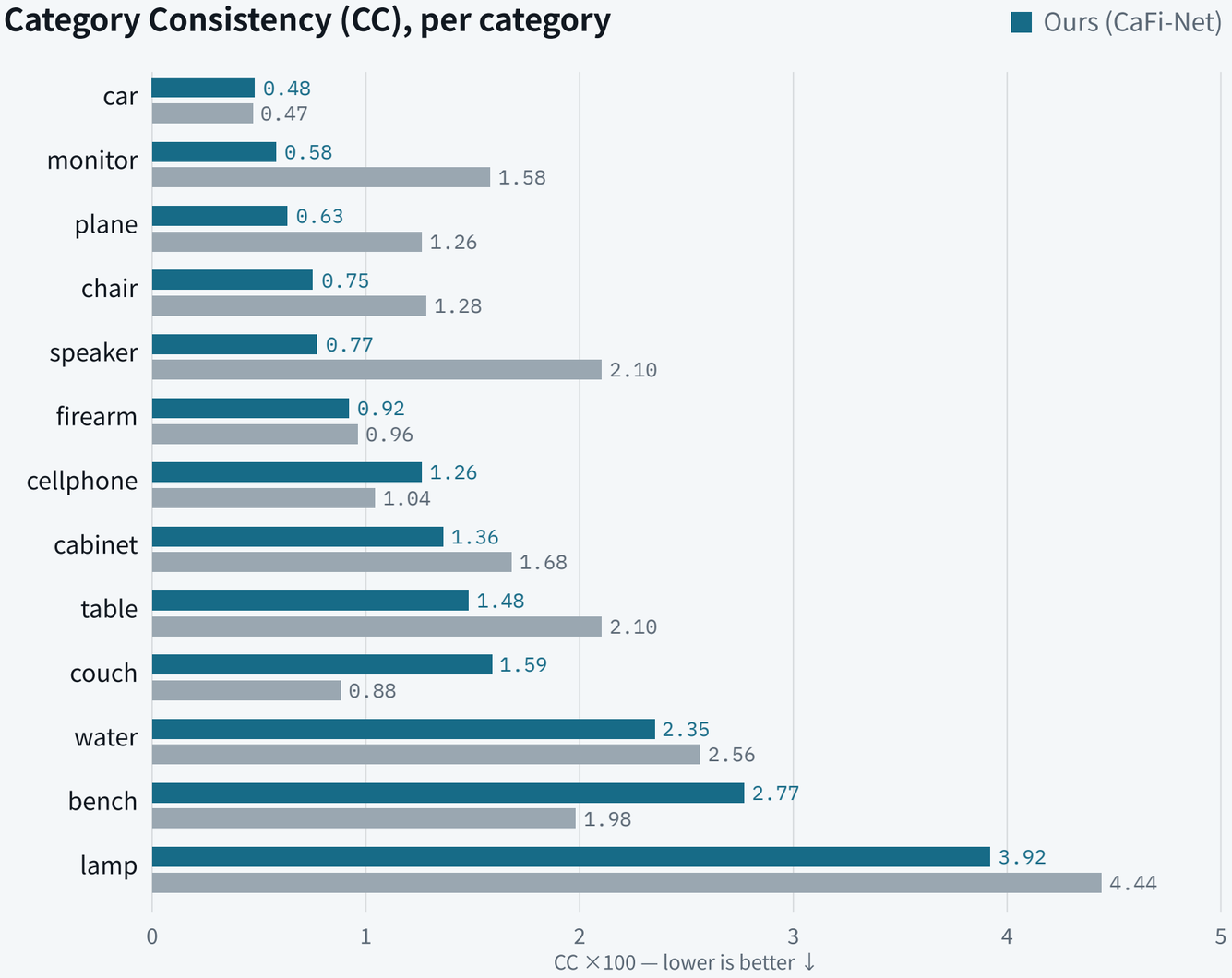
TAKEAWAY CaFi-Net achieves the **best CC and GEC** among all methods and is **on par with ConDor on IC** — while operating directly on the density field, not point clouds.

Source: Canonical Fields (CVPR 2023), p. 6, Table 1 · p. 7, §5

RESULTS · PER-CATEGORY ANALYSIS

Competitive Across All 13 Categories

Category Consistency (CC), per category



Source: Canonical Fields (CVPR 2023), p. 6, Table 1

FINDINGS

Low variance across categories — CaFi-Net is competitive everywhere, with no category where it clearly loses.

p. 6, §5.1

Firearms and lamps show similar trends across metrics — category shape plays a role in canonicalization difficulty.

p. 6, §5.1

PCA is unreliable on noisy point clouds; CaCa fails on high-noise categories. ConDor's TFN choice adds noise robustness.

p. 7, §5.1

ABLATION STUDIES

Each Design Choice Earns Its Place

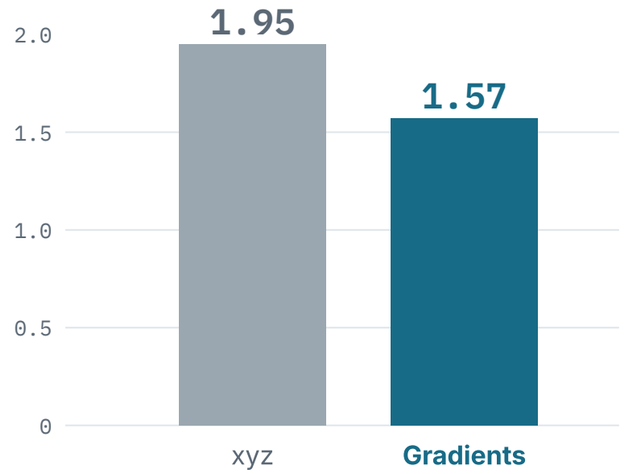
Three ablations on 4 categories (bench, cellphone, chair, airplane) — GEC average, ×100, lower is better.

Ablated variant Full CaFi-Net

GEC ×100 · lower is better

(a) xyz → Gradients

Signal representation: raw coordinates vs density gradients

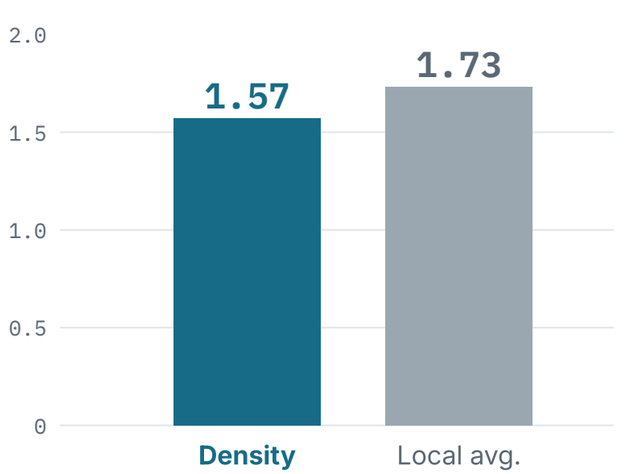


Gradients weigh features **near object surfaces higher** → better canonicalization.

IC 1.73 → 1.26 · CC 1.72 → 1.35

(b) Density → Local Avg.

Weighting: direct density vs local average density

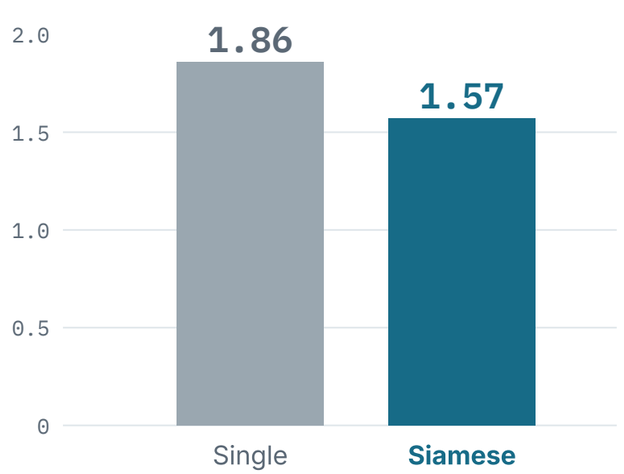


Local averaging **smooths out details** → direct density wins.

IC 1.26 → 1.85 · CC 1.35 → 1.49

(c) Single → Siamese

Architecture: single branch vs Siamese network



Siamese branch **improves all metrics** (IC, CC, GEC).

IC 1.48 → 1.26 · CC 1.62 → 1.35

Source: Canonical Fields (CVPR 2023), p. 7, Table 2

VISUAL ANALYSIS

Qualitative Comparison & Failure Cases

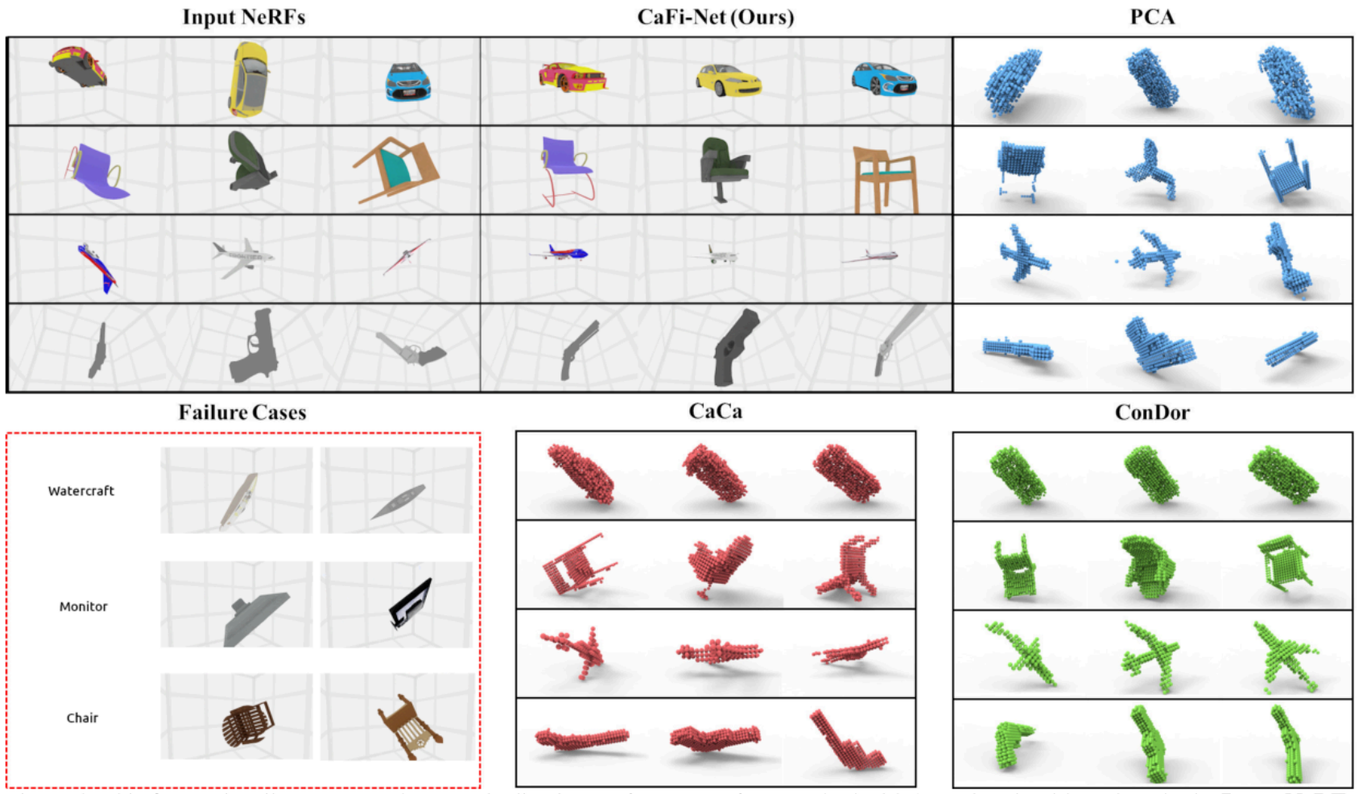


Figure 4. This figure visually compares the canonicalization performance of our method with 3D point cloud-based methods. **Input NeRFs** shows one of the RGB views used to learn our NeRF input (notice how orientations of instances are random). **CaFi-Net (Ours)** shows the results of our canonical field rendered from a novel view unseen during NeRF training. **PCA** (blue), **CaCa** [47] (red), and **ConDor** [37]'s (green) results are also presented for the same instances. Our method matches or exceeds ConDor. We also show some failure cases for several instances (red box) which occur either due to thin structures or symmetry.

Novel-view render

"Ours" columns render the canonical field from a view **unseen during NeRF training** — not a memorized input view.

Baseline color coding

Point-cloud baselines: ■ PCA, ■ CaCa, ■ ConDor — same instances as CaFi-Net.

Failure cases

Watercraft, monitor, chair (red dashed box): thin structures or symmetry defeat canonicalization.

Source: Canonical Fields (CVPR 2023), p. 8, Figure 4

KEY TAKEAWAYS & LIMITATIONS

What CaFi-Net Delivers — and Where It Breaks

Three contributions, three honest limitations. No hedging.

What we delivered CONTRIBUTIONS · p. 2

✓ **First self-supervised field canonicalizer**
First method to canonicalize 3D pose of objects represented as neural radiance fields — directly on the field, no point clouds.

✓ **Siamese equivariant architecture**
Siamese network with equivariant feature extraction operating directly on continuous, noisy NeRF density fields.

✓ **Public dataset**
1300 NeRF models across 13 ShapeNet categories — posed images and trained weights released.

Where it breaks LIMITATIONS · p. 8

! **Requires 360° NeRFs**
Cannot handle partial views from front-facing NeRFs — full omnidirectional coverage is assumed.

! **Thin / small structures**
Density uncertainty grows for very thin or small structures that can be missed (chair case in Figure 4).

! **TFN symmetry issues**
Inherits Tensor Field Network problems with symmetric objects (monitor case in Figure 4).

Source: Canonical Fields (CVPR 2023), pp. 2, 8

CONCLUSION

From arbitrary poses to one canonical frame.

SUMMARY

- CaFi-Net self-supervisedly canonicalizes the 3D pose of objects represented as neural radiance fields — directly on the noisy, continuous density field, with no point clouds.
- A Siamese network with rotation-equivariant convolution layers, trained on 1300 NeRF models across 13 ShapeNet categories.
- Matches or outperforms 3D point-cloud-based canonicalization methods.

FUTURE WORK

- Build a large real dataset of common object categories without manual pose canonicalization.
- Combat the symmetry issue.

Source: Canonical Fields (CVPR 2023), p. 8

APPENDIX

Key References

[5] Chang et al., ShapeNet, 2015.

[9] Deng et al., Vector Neurons, ICCV 2021.

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[23] Mildenhall et al., NeRF in the Dark, CVPR 2022.

[24] Mildenhall et al., NeRF: Representing Scenes as Neural Radiance Fields, ECCV 2020.

[32] Karl Pearson, PCA, 1901.

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[37] Sajnani et al., ConDor, CVPR 2022.

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[50] Thomas et al., Tensor Field Networks, 2018.

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Thank you.

Canonical Fields: Self-Supervised Learning of Pose-Canonicalized Neural Fields

Shaurya Dewan[1] • Rahul Sajnani[2] • Adrien Poulenard[3] • Rohith Agaram[1] • Madhava Krishna[1] • Srinath Sridhar[2]
IIIT-Hyderabad • Brown University • Stanford University

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ivl.cs.brown.edu/projects/canonicalfields